Soroush Mahdi

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Education

Master of Science in Artificial Intelligence — *Tehran, Iran*
AmirKabir University of Technology (Tehran Polytechnic) — *Sep 2020 – Sep 2023*
Ranked ***7th*** *among 50 M.Sc. students*
Cumulative ***GPA: 4/4***

Bachelor of Computer Engineering — *Hamedan, Iran*
Bu-Ali Sina University — *Sep 2016 – Sep 2020*
Cumulative GPA: 3.17

Research Interests

Vision-Language Models (VLMs)	**Computer Vision**
Computational Neuroscience	**Deep Learning Theory**
Trustworthy ML	**Generative AI**

Publications

1. **Mahdi, S.**, Ayar, F., Javanmardi, E., Tsukada, M., Javanmardi, M., "Evict3R: Training-Free Token Eviction for Memory-Bounded Streaming Visual Geometry Transformers". Submitted to ICRA 2025.
2. **Mahdi, S.**, Nabati, S., Dehghanian, Z., Amirmazlaghani, M., "MemLoss: Enhancing Adversarial Training with Momentum-Based Memory". Submitted to Applied Soft Computing.

Honors & Awards

- **1st** Place in The Iran Annual AI Award, **focusing on Brain abnormality diagnosis using MRI DICOM images**, Cognitive Sciences and Technologies Council, Tehran, Iran, 2024.
- **2nd** Place in The National fNIRS Data Analysis Competition focusing on brain-computer interface systems to help patients with movement disorders, National Brain Mapping Laboratory, Tehran, Iran, 2022.
- Ranked **61st Among More Than 15,000 Participants** in Iranian University Entrance Exam For Masters In Artificial Intelligence, Iran, 2020.
- Ranked **36th Among More Than 15,000 Participants** in Iranian University Entrance Exam For Masters In Algorithms and Computation, Iran, 2020.
- Ranked **22nd** Place in National Collegiate Scientific Olympiad in Computer Engineering, Iran National Organization of Educational Testing, Tehran, Iran, 2020.

Academic Experiences

Research Assistant — *Jun 2025 – Present*
Autonomous & Intelligent Systems Lab (AISL), Amirkabir University of Technology — *Tehran, Iran*
Under the supervision of Mahdi Javanmardi

- Working on 3D Computer Vision and Vision Language Models.
- Developed Evict3r, a novel token eviction method for deep foundation SFM models, improving efficiency in large-scale scene reconstruction and reducing computational overhead.

Thesis Based M.Sc. Student and Research Assistant — *June 2021 – Present*
Statistical Data Analysis Laboratory — *Tehran, Iran*
Under the supervision of Maryam Amirmazlaghani

- Thesis title: An approach for improving the robustness of deep neural network image classifiers against adversarial examples.

Work Experiences

Data Scientist — *July 2024 – June 2025*
Iran Credit Scoring — *Tehran, Iran*
•Working as a data scientist in the R&D team.

Computer Vision Researcher — *Jan 2022 – Jan 2023*
HARA.ai — *Tehran, Iran*
•Specialized in face anti-spoofing and liveness detection using deep learning techniques.
•Developed blink detection and blink counter models using semi-supervised approaches.

Academic Projects

- **Robust Supervised contrastive learning** - Github link
  Implemented adversarial training coupled with supervised contrastive learning using PyTorch.
- **PSO-SGD Hybrid Optimizer** - Github link
  Developed a custom PyTorch optimizer by designing an optimization algorithm that combines SGD and PSO.
- **multivariate HMM** - Github link
  Implemented a range of algorithms for Hidden Markov Models (HHMs) in the multivariate case from scratch.

Teaching Assistantships

Statistical Machine Learning - AmirKabir University of Technology — *Tehran, Iran*
Instructor: Maryam Amir mazlaghani (Assoc. Prof.) - Graduate Course — *Spring 2023*

Machine Vision - AmirKabir University of Technology — *Tehran, Iran*
Instructor: Reza Safabakhsh (Prof.) - Graduate Course — *Fall 2022*

Other Courses: Algorithm Design(2020), Artificial intelligence & Expert Systems(2019), Data Structures(2018)

Professional Skills

- **Programming Languages**:
  Python, C++, R
- **Libraries/Frameworks**:
  PyTorch, NumPy, Pandas, OpenCV, scikit-learn, Tensorflow, JAX
- **Tools**:
  Git, LATEX, Visual Studio Code
- **Languages**:
  English: TOEFL iBT Score: 103/120 (Reading: 30/30, Listening: 29/30, Speaking: 21/30, Writing: 23/30)
  Persian: Native

Major Courses

Machine Vision	19.9/20	**Stochastic Processes**	19.28/20
Machine Learning	18.6/20	**Neural Networks**	16.8/20
Optimization	16.21/20	**Computational Intelligence**	20/20

References

Mahdi Javanmardi (Asst. prof.)	mjavan@aut.ac.ir	Homepage
Maryam Amir Mazlaghani (Assoc. Prof.)	mazlaghani@aut.ac.ir	Homepage
Mir Hossein Dezfoulian (Asst. Prof.)	dezfoulian@basu.ac.ir	Homepage

More references are available upon request.